FIRMGENIE: A CONFIDENCE-AWARE EXPERT LLM FRAMEWORK FOR FIRMWARE VERSION IDENTIFICATION ON NETWORK DEVICES

Fengshi Zhang[1,2], *Zhi Li*[1,2,†], *Shunchao Xu*[1,2], *Zekun Zhang*[1,2],
Nan Yu[1,2], *Yongle Chen*[3], *Dongliang Fang*[1,2], *Limin Sun*[1,2]

[1]Beijing Key Laboratory of IOT Information Security Technology, Institute of Information Engineering, CAS, Beijing, China
[2]School of Cyber Security, University of Chinese Academy of Sciences, Beijing, China
[3]College of Computer Science and Technology, Taiyuan University of Technology, Taiyuan, China

ABSTRACT

The explosive growth of Internet of Things (IoT) devices makes firmware-level identification vital for cybersecurity, yet existing methods fail to scale and struggle with noisy, unstructured data. We propose *FirmGenie*, the first expert LLM framework for robust, large-scale firmware version identification. Our core insight is that a compact, specialized LLM, developed through knowledge distillation and domain-specific fine-tuning, can significantly outperform much larger general-purpose models while utilizing only approximately 1% of the parameters of state-of-the-art models like DeepSeek-R1 (671B). This drastic reduction in model size ensures high inference efficiency and deployability for real-time network analysis. To ensure reliability, *FirmGenie* employs a knowledge augmentation mechanism and a novel multi-source confidence discriminator to enhance inference and suppress false positives. Our method achieves state-of-the-art results of firmware version accuracy exceeding 96%, outperforming both general-purpose LLMs and traditional scanning methodologies.

Index Terms— IoT Security, Firmware Version Identification, IoT Device Identification, Large Language Model

1. INTRODUCTION

The rapid proliferation of Internet of Things (IoT) devices is reshaping modern networks while introducing severe security risks, making large-scale, high-accuracy device identification a cornerstone of cybersecurity. However, current identification techniques face a **granularity bottleneck**: most stop at recognizing the device type, brand, or model [1–3]. This level of detail is insufficient for precise risk assessment, as the exploitation and remediation of security vulnerabilities are directly tied to **specific firmware versions** [4, 5]. Addressing this limitation requires more fine-grained identification.

Existing fine-grained device identification methods fall into two categories. The first category is based on explicit information, with existing methods employing handcrafted rules [6, 7] or static fingerprint databases [8–10]. While straightforward for known devices, these methods are labor-intensive to maintain [11], difficult to scale within the dynamic IoT ecosystem [12], and prone to false positives from noisy data [13]. The second category is based on implicit-features, including machine-learning–based firmware feature extraction [14, 15] and firmware file hash matching [16, 17]. However, these methods are often impractical for large-scale, real-time deployment due to their complex feature engineering and reliance on firmware images [7]. Consequently, achieving scalable and accurate firmware version identification remains a significant challenge.

To address these bottlenecks, we explore the potential of large language models (LLMs), which have achieved remarkable success in natural language processing [18, 19]. Existing LLM-based approaches for device identification [20, 21] have not tackled the much harder problem of firmware version identification, which is uniquely challenging because this task requires reasoning over scarce labeled data and inconsistent network banner patterns. Thus, we propose to leverage the strong inference and generative capabilities of LLMs, which allow them to reason about novel, unseen data, a critical requirement in a data-scarce environment. However, using LLMs for this task introduces three key challenges: (1) the trade-off between the high cost of large models and the weaker performance of small ones; (2) their inherent sensitivity to input formatting and lack of robustness in reasoning over specialized, semi-structured data [12], such as the highly variable patterns found in network banners; and (3) a high false-positive rate on real-world negative samples [13].

To solve these challenges, we propose ***FirmGenie***, the first expert LLM framework for large-scale, fine-grained IoT device firmware version identification. The core of ***FirmGenie*** is a compact expert LLM, which balances deployment efficiency with specialized reasoning through Supervised Fine-Tuning (SFT) on domain data and knowledge distillation.

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To ensure accuracy and robustness, ***FirmGenie*** employs a knowledge augmentation mechanism validates and enhances model inferences against a comprehensive device database, improving performance on unseen devices. Subsequently, a multi-source confidence discriminator provides the final reliability check, suppressing false positives by evaluating model Chain-of-Thoughts (CoTs) and multi-stage confidence scores. Finally, ***FirmGenie*** output the best Brand, Model, Firmware Version (B, M, V) candidate. Experimental results demonstrate that ***FirmGenie*** effectively addresses these challenges and achieves state-of-the-art performance. Our framework is available at: https://github.com/lululu930/FirmGenie

Our main contributions are as follows:

- We present *FirmGenie*, the first expert LLM framework to achieve firmware version identification at internet scale, overcoming the granularity limits of prior device identification methods and the key domain adaptation challenges for LLMs (Targeting challenge (1) & (2)).
- We design a multi-source confidence discriminator that evaluates internal reasoning states to suppress false positives from negative samples, ensuring high reliability in real-world scenarios (Targeting challenge (3)).
- We achieve state-of-the-art results on real-world data, with firmware version accuracy exceeding 96% and F1-score exceeding 97%, outperforming both general-purpose LLMs and traditional scanning methodologies.

2. METHODOLOGY

2.1. Framework Overview

Figure 1 presents the overall framework of *FirmGenie*, which integrates domain knowledge and large language models for accurate firmware identification. At its core is *SFT-Qwen*, an expert model obtained through domain-specific supervised fine-tuning of Qwen2.5-7b [22] with knowledge distilled from DeepSeek-R1 [19]. SFT-Qwen firstly generating candidate brand, model, and firmware version keywords with confidence scores. Then, candidates are verified through a domain-specific knowledge graph, the related knowledge are fed into SFT-Qwen for a second reasoning round. Finally, a pre-trained confidence discriminator fuses multi-dimensional confidence signals and CoT semantics to evaluate the outputs.

2.2. Candidate Generation via Expert LLM

Knowledge Distillation: To balance the reasoning ability of large LLMs with the efficiency of smaller ones, we employ knowledge distillation to transfer the reasoning capabilities of DeepSeek-R1 to Qwen2.5-7b. Labeled banners are fed to DeepSeek-R1 with carefully designed prompts (with expert knowledge on how to score the keywords) to produce detailed CoTs and confidence scores for candidate extraction.

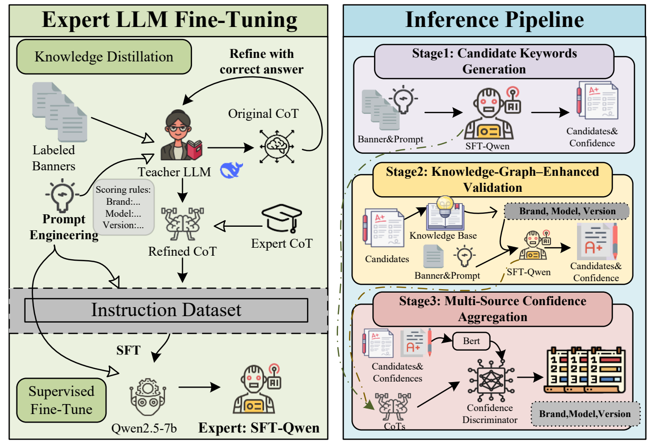

Fig. 1: Overview of *FirmGenie*. The left part presents our expert LLM fine-tuning workflow, the right part illustrates our firmware version identification pipeline, which consists of three stages.

CoT Refinement: We observed that DeepSeek-R1 often produced verbose and redundant CoT for banners lacking direct evidence, which is inefficient for training smaller models. To address this, we prompted DeepSeek-R1 to self-refine the CoT of correctly identified samples, pruning them to the most essential logic. Additionally, for hard samples where DeepSeek-R1 failed, we manually crafted expert CoTs. This dual strategy significantly improved the training efficiency and learning outcomes of student model.

SFT-Qwen: Using the refined CoTs, we conduct SFT on Qwen2.5-7b and get an expert LLM, denoted as SFT-Qwen.

Candidate Keyword Generation: For $Banner_k$, we first apply a preprocessing step to clean and segment long text. The processed text is then fed to SFT-Qwen with a designed prompt. The output is structured as: $\{Attribute : [(c_i, s_i)]\}$ where $Attribute \in$ {brand, model, firmware version}, c_i is the i^{th} candidate keyword, $s_i \in [0, 1]$ is the confidence score.

2.3. Knowledge-Augmented Validation

Due to the complexity and noise of banners, SFT-Qwen may generate multiple candidates, but its internal knowledge struggles to cover the vast relationships of (B, M, V) across the entire IoT ecosystem. To break this limitation, we designed a knowledge-augmented validation mechanism to enhance and validate the outputs of SFT-Qwen.

KG Construction: We constructed a domain-specific knowledge graph (KG) of (B, M, V) relations aggregated from multiple sources (Section 3.1), enabling real-time retrieval to feed the results back into SFT-Qwen for enhanced reasoning.

KG-Enhanced Validation: We calculate a multi-factor weighted similarity score $Sim_{i,j}$ based on: (1) Numerical difference (2) Lexical Jaccard similarity (3) Edit distance similarity. (4) Bonus for prefixes, suffixes, and containment. i, j denotes the candidate keyword and the KG keyword. Due to space constraints, this will not be elaborated here. The validation is a two-step process: first, we attempt a direct match of existing tuples in the KG. If that fails, we provide SFT-Qwen with a list of all (B, M, V) relationships from the KG keywords that have $Sim_{i,j}$ above the defined threshold to the candidate keyword for a second identification round.

2.4. Multi-Source Confidence Aggregation

Relying on a single confidence signal may insufficient. The confidence of LLM can suffer from hallucinations and over-confidence [23], if the model lacks knowledge of vendor-specific naming conventions, it may assign high confidence to incorrect keywords. Conversely, while confidence from KG is generally reliable, an incomplete relation may lead the model to assign low score to correct but unlisted keyword.

The CoTs of LLM contains rich semantic cues that reveal the LLM decision process [24]. In case of negative samples, when a banner lacks explicit evidence for an extracted answer, the model's reasoning process tends toward speculation and hesitation (e.g., using words like "possibly"). However, such semantic signals are difficult to quantify during inference. To address these issues, we design a Multi-Source Confidence Discriminator (CD) that trained on three features: (1) The LLM confidence score C_{LLM} produced by candidate keyword generation; (2) The LLM confidence score C_{KB} obtained during knowledge-augmented validation; (3) The CoTs and candidate keyword generated by the LLM across these stages, which are encoded into semantic representation vectors h(T) using a pre-trained BERT model. The concatenated features form a joint representation that is fed into our CD, which then outputs the prediction probability as:

$$\hat{y} = \sigma\big(W_2\, \phi\big(W_1[\mathbf{h}(T) \oplus C_{\text{LLM}} \oplus C_{\text{KB}}] + b_1\big) + b_2\big) \qquad (1)$$

this probability becomes the final score of each keyword.

By modeling confidence signals from multi-stages with CoT semantic representations, the CD mitigating biases from single source, improving overall robustness and reliability.

3. EXPERIMENTS

3.1. Datasets

Our dataset consists of three components: a device information knowledge base, a labeled ground-truth set, and a unlabeled in-the-wild dataset. Data details are listed in Table 1.

Domain-specific KG Database: Information on (B, M, V) relations was extensively collected through (1) the CPE [5] database; (2) device database from network asset mapping platform Cydar [25]; and (3) firmware data scraped from 14 vendor websites. This database is designed to be continuously updated to support the identification of new devices.

Labeled Dataset: We scanned over 100 million public IPs [10, 25, 26] on approximately 1,000 of the most common TCP ports. After initial filtering and annotating by Cydar [25], each host was manually labeled (B, M, V) by three IoT security experts, with a fourth resolving any disagreements. The difficulty of fine-grained version identification limits dataset scale, as large-scale automated labeling is ineffective and manual annotation is impractical. We curate a 5:1 positive-to-negative ratio (negatives lack firmware version information, with no constraints on brand or model). The data are initially split 2:3 for training and testing; high-quality CoT samples from DeepSeek-R1 are selected from the training set for SFT and confidence discriminator training, and the remaining data are merged into the test set.

Table 1: Details of Dataset Statistics

Category	Split	Samples	B	M	V	P
KG	CPE	-	1.3k	10k	41k	-
	Cydar	-	1.6k	104k	267k	-
	Crawled	-	14	16k	180k	-
	Total	-	2.7k	119k	256k	-
Labeled	SFT	1516	745	1391	1142	35
	Test	4000	1345	2445	1892	46
	Total	5516	1451	3251	2802	46
Unlabeled	-	150k	-	-	-	94

Note: B,M,V,P denote Brand, Model, Firmware Version, and Protocol.

Unlabeled Dataset: In-the-Wild dataset, collected in the same manner as the labeled dataset, initially filtered by Cydar to remove non-IoT-related data, remains unverified and unlabeled, where the data distribution is close to the real-world ratio of positive and negative samples in cyberspace.

3.2. Performance Evaluation

Experimental Settings: (1) SFT-Qwen: We performed SFT on Qwen2.5-7b using LoRA [27] with a learning rate of 5e-5 on four NVIDIA V100 GPUs. (2) Confidence Discriminator: We utilize BERT as the backbone encoder, trained on a Two-layer MLP with 128 hidden units and ReLU activation.

Baselines: (1) Nmap [8]: use handcrafted rules and fingerprint databases. Serve as a benchmark for traditional expert-driven approaches. (2) Shodan [10]: Serve as a benchmark of device search engine. (3) DeepSeek-R1, Qwen-14b [28], Qwen2.5-7b, Llama3-8b [29]: state-of-the-art LLMs, establishing the baseline of powerful, general-purpose models.

Evaluation Metrics: We use the Precision, Recall, F1-score, False Positive Rate (FPR) to evaluate our performance:

$$\text{Precision} = \frac{TP}{TP + FP} \qquad \text{Recall} = \frac{TP}{TP + FN}$$
$$\text{FPR} = \frac{FP}{FP + TN} \qquad \text{F1} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \qquad (2)$$

where TP, FP, TN and FN refer to the true positive, false positive, true negative and false negative. To comprehensively evaluate the performance on both positive and negative samples, we calculate Precision, Recall, and F1-score on positive samples, and evaluate the FPR on negative samples.

Table 2: Performance Comparison of State-of-the-Art Firmware Version Identification Methods

Method	Firmware Version		
	Precision	Recall	F1
Nmap	1.000	0.001	0.003
Shodan	0.606	0.033	0.062
Qwen2.5-7b	0.798	0.842	0.819
LLaMA-8b	0.872	0.907	0.889
Qwen-14b	0.909	0.967	0.937
FirmGenie (7b)	**0.963**	**0.985**	**0.977**
DeepSeek-R1	0.964	0.980	0.972

Results: As detailed in Table 2, *FirmGenie* demonstrates superior firmware version identification performance, achieving an F1-score over 325× higher than Nmap and 15× higher than Shodan, which highlights the limitations of traditional tools. While Nmap yields perfect precision by extracting known versions from specific formats using static rules, its negligible recall renders it unsuitable for the diverse IoT ecosystem.

Through knowledge augmentation and multi-source confidence control, *FirmGenie* outperforms LLMs across parameter scales and matches or slightly exceeds its teacher model DeepSeek-R1, despite using only about 1% of its parameters, and the architecture can flexibly scale to larger or smaller models as needed. This demonstrates that *FirmGenie* can overcome the inherent limitations of traditional methods and general LLMs in firmware version identification, setting a new, efficient benchmark for IoT device firmware version identification at internet scale. The identified firmware directly support downstream tasks such as asset mapping and vulnerability matching, strengthening overall IoT and cyberspace security.

In fact, *FirmGenie* also supports brand and model identification; however, these are only auxiliary, coarse-grained steps and are not our primary optimization focus, so their performance is omitted. *FirmGenie* remains effective even when banners contain only firmware-related information.

Large-scale analysis: As shown in Figure 2, the analysis on our unlabeled dataset reveals that *FirmGenie* identified 129,268 devices in total (identified either the brand/model or firmware version). Crucially, it identified firmware versions for 58,451 of these devices, surpassing Nmap by 211 times (58,451 vs. 277), highlights the profound superiority and potential of *FirmGenie* in large-scale firmware security analysis.

3.3. Ablation Study

We conducted ablation studies to validate each component of *FirmGenie*, with results shown in Table 3. Our SFT strategy significantly improved performance across all metrics, confirming the value of domain-specific SFT. Integrating the knowledge augment for second round extraction enhanced performance across all metrics, pushing the precision to its peak of 0.978 while effectively reducing the FPR to 0.18. Finally, our multi-source CD further balance the recall-precision trade-off, maximizing the recall to 0.985 and cutting the FPR to 0.09 with a minor precision decrease (0.015). This occurs because CD considers the scores in multiple stages, may filter uncertain but correct answers, which is within acceptable bounds and provides a crucial benefit: the threshold of CD can be adjusted based on the specific application scenario, allowing operators to choose their desired balance between security (fewer FPs) and coverage.

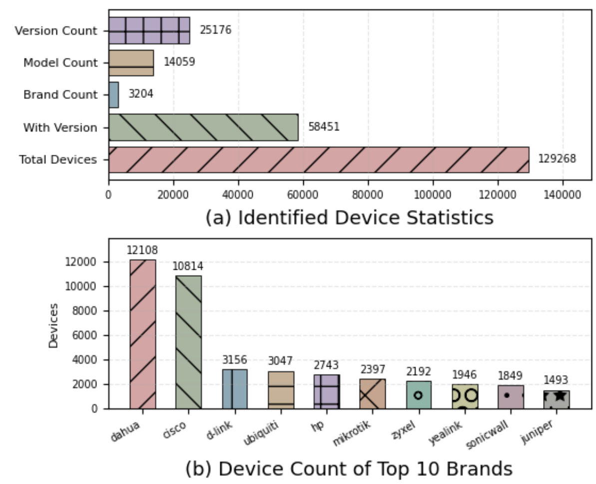

Fig. 2: Identification Results on Unlabeled Data.

Table 3: Results of the Ablation Study

Method	Firmware Version			
	Precision	Recall	F1	FPR
Qwen2.5-7b	0.798	0.842	0.819	0.49
SFT-Qwen2.5-7b	0.947	0.885	0.915	0.32
SFT-Qwen2.5-7b + KG	**0.978**	0.906	0.941	0.18
FirmGenie (CT=0.9)	0.963	**0.985**	**0.977**	**0.09**

Note: "CT" means confidence threshold of confidence discriminator.

4. CONCLUSION

We propose *FirmGenie*, an expert LLM framework addressing the challenge of IoT firmware identification. By integrating domain-specific SFT, knowledge-augmented validation, and a multi-source confidence discriminator, *FirmGenie* overcomes the limitations of traditional methods and general LLMs to achieve unprecedented accuracy and scale. Experiments show it achieves an F1-score of 0.977, surpassing traditional methods and general LLMs using only 1% of the parameters compared to DeepSeek-R1. Moreover, *FirmGenie* identifies 211x more versions than traditional methods in large-scale analysis, establishes a new, efficient benchmark for large-scale, automated firmware identification.

A. MODEL EFFICIENCY ANALYSIS

This appendix presents additional experiments added during the rebuttal phase in response to reviewers' requests for clearer evidence on model efficiency, inference cost, and system-level behavior. These results complement the main evaluation and further assess the deployability of *FirmGenie*.

A.1. Model Efficiency and Cost Analysis

Table 4 compares inference latency, throughput, and operating cost across representative LLM baselines and *FirmGenie*. Despite using only a 7B backbone, *FirmGenie* achieves the lowest end-to-end latency among all evaluated systems and processes 272 samples per hour for the full pipeline. When considering only the expert LLM component (SFT-Qwen-7B), throughput increases to 461 samples per hour, substantially outperforming larger general-purpose models. In contrast, DeepSeek-R1 (671B) exhibits the highest latency, lowest throughput, and incurs non-negligible API costs, whereas *FirmGenie* supports zero-cost local deployment. Overall, these results demonstrate that domain specialization enables a favorable accuracy–efficiency trade-off.

Table 4: Inference Efficiency and Operating Cost Comparison Across Models

Model	Latency (s)	Samples/h	Cost ($)
Qwen2.5-7B	16.978	212	0.00
Qwen-14B	13.467	267	0.00
LLaMA-8B	14.266	252	0.00
DeepSeek-R1 (671B)	21.544	167	3.21
SFT-Qwen-7B (Ours)	7.815	461	0.00
Full Pipeline (Ours)	**13.249**	**272**	**0.00**

A.2. Latency and Throughput Breakdown

Figures 3 and 4 visualize the latency and throughput trends across models. While scaling model size (e.g., Qwen-14B, LLaMA-8B) yields moderate efficiency gains over Qwen2.5-7B, these models remain significantly slower than *FirmGenie*. This indicates that efficiency improvements stem primarily from task-specific design and confidence-aware filtering, rather than parameter scaling alone.

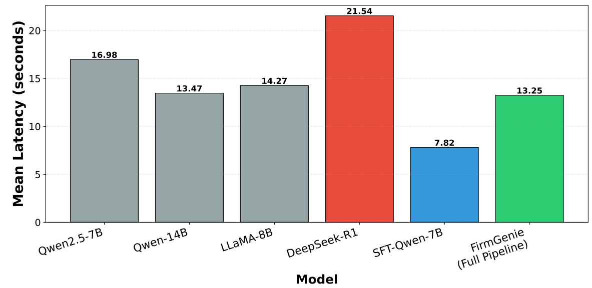

Fig. 3: End-to-End Inference Latency Comparison.

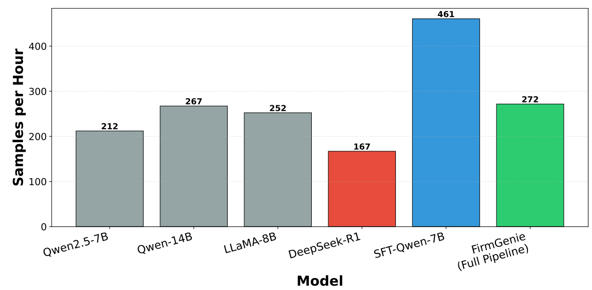

Fig. 4: Inference Throughput Comparison (Samples per Hour).

A.3. Pipeline-Level Analysis

Figure 5 decomposes the runtime of *FirmGenie* into its constituent stages. LLM inference accounts for 62.4% of total runtime, followed by knowledge-based validation (37.3%), while the confidence discriminator contributes negligible overhead (0.3%). This confirms that the proposed confidence control mechanism improves reliability without materially impacting system efficiency, supporting large-scale deployment.

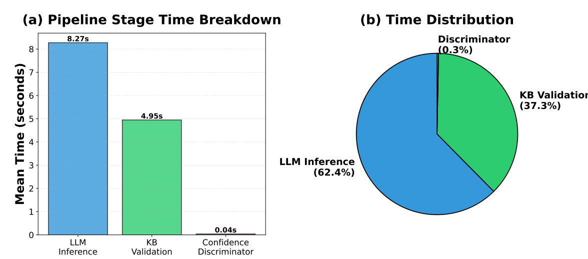

Fig. 5: Runtime Breakdown of the FirmGenie Pipeline.

Together, these experiments demonstrate that *FirmGenie* is not only accurate but also operationally efficient. The results reinforce our central claim that a compact, expert LLM augmented with structured knowledge and confidence control can outperform much larger models while remaining practical for Internet-scale firmware version identification.

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